

Wrexham County Borough Council Section 19 Report

Worthenbury Community Flood Event: 16th October 2024

S19 Flood Investigation Report

This document is available in Welsh

Document Control Sheet

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Approval

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1. Introduction

As a Lead Local Flood Authority (LLFA), it is the responsibility of Wrexham County Borough Council (WCBC) to manage the local flood risk from surface water, groundwater and ordinary watercourses such as streams and ditches (including lakes and ponds). The following report has been prepared in accordance with the requirements of the Section 19 of the Flood and Water Management Act 2010 (FWMA) and WCBCs Local Flood Risk Management Strategy (LFRMS) and Plan (LFRMP).

This investigation has been carried out following recent internal flooding of residential properties.

In accordance with the requirements of Section 19 of the FWMA (2010):

- 1) *On becoming aware of a flood in its area, a lead local flood authority must, to the extent that it considers it necessary or appropriate, investigate-*
 - a) *which risk management authorities had relevant flood risk management functions; and*
 - b) *whether each of those risk management authorities had exercised or was proposing to exercise those functions in response to the flood.*
- 2) *where an authority carries out an investigation under subsection (1) it must-*
 - a) *publish the results of its investigation, and*
 - b) *notify any relevant risk management authorities.*

2. Flood Chronology

2.1 Recent Events

On Wednesday 16th October 2024, a band of high intensity rainfall travelled northwards across the east of the county borough. This stalled over the area dropping approximately 63.2 mm of rainfall across the Wych Brook and Emral Brook upstream of Worthenbury over a period of ten hours (Met Office, 2025). This caused the water level in Worthenbury Brook to rise to 3.422 m before the data logger failed from inundation. The failure occurred because the logger is designed to monitor low water levels for abstraction downstream. However, given that the flood embankment was exceeded in locations where the height of the flood embankment had settled, it is estimated the flood peak could have been as high as 14.0 mAOD.

An operational response was instigated at 13:45 allowing for NRW officers to attend Worthenbury from 15:00 on the afternoon of 16th October. At approximately 21:00 it was reported to Natural Resources Wales that the flood embankment defense protecting The Cobbles, Quinton, The Manor and Manor Cottage was overtopping into the defended area on the southern, landward side of the defense. The latter three of these properties form separate parts of a Grade II Listed Building. By the time that the overtopping had been observed, water ingress under the flood embankment had also been noted by residents and officers from NRW in an area to the south east of the defended area. It is understood by the LLFA that this problem was existing from a high flow event in September 2024, and NRW had mitigated this through the placing and securing of a visqueen cover until a more permanent repair could be made. In addition, photographic evidence provided by NRW shows flood water overtopping on to the public highway which then found its way into the driveways of the properties.

All four properties within the defended area subsequently started to flood around 22:00 and had reportedly reached a height of 0.91 m inside one of the properties by 03:00 on 17th October 2025. The flood water had to be pumped into the river over the next three days following the event.

2.2 Flood History

The village of Worthenbury suffers frequent flooding from Emral, Wych and Worthenbury Brook due to the large upstream catchment area. The proximity of the village to the confluence of Worthenbury Brook and the River Dee also means that water levels are frequently raised immediately downstream of the village. It is very common for the village to become cut off following flood events as flood waters often block the B5069.

The flood problem surrounding the properties covered in this report are highway surface water runoff, surcharged land drains and ordinary watercourses. During larger floods there is major inundation of the B5069 into the village, such as occurred in October and November 2000. The flooding of these properties has been documented for the past thirty years. The property known as The Cobbles has been recorded as suffering from flooding in February 1990, March 1991, February 1994, March 1995, winters 1998 and 1999. The properties known as Quinton, The Manor and Manor Cottage having their access cut off during these winters as well. During the floods of October and November 2000 all four properties suffered twice from major flooding.

The above properties have been protected by a horseshoe shaped flood defense which was constructed in 2004. However, it should be noted that The Cobbles regularly suffers from groundwater flooding issues. Bow waves from the flooded highway also contribute to the flooding of this property.

3. Current Management Regime

The flood defenses which protect the houses that were most impacted by flooding are maintained by Natural Resources Wales (NRW) who are the risk management authority for main river flooding. A range of landowners have riparian ownership over both the river and an ordinary watercourse which interacts with the floodplain. Wrexham County Borough Council are responsible for the highway and the culverts which runs underneath it.



Figure 1: A map showing the location of the flood embankment maintained by Natural Resources Wales (NRW) taken from the Flood Defence Structures map on Data Map Wales.

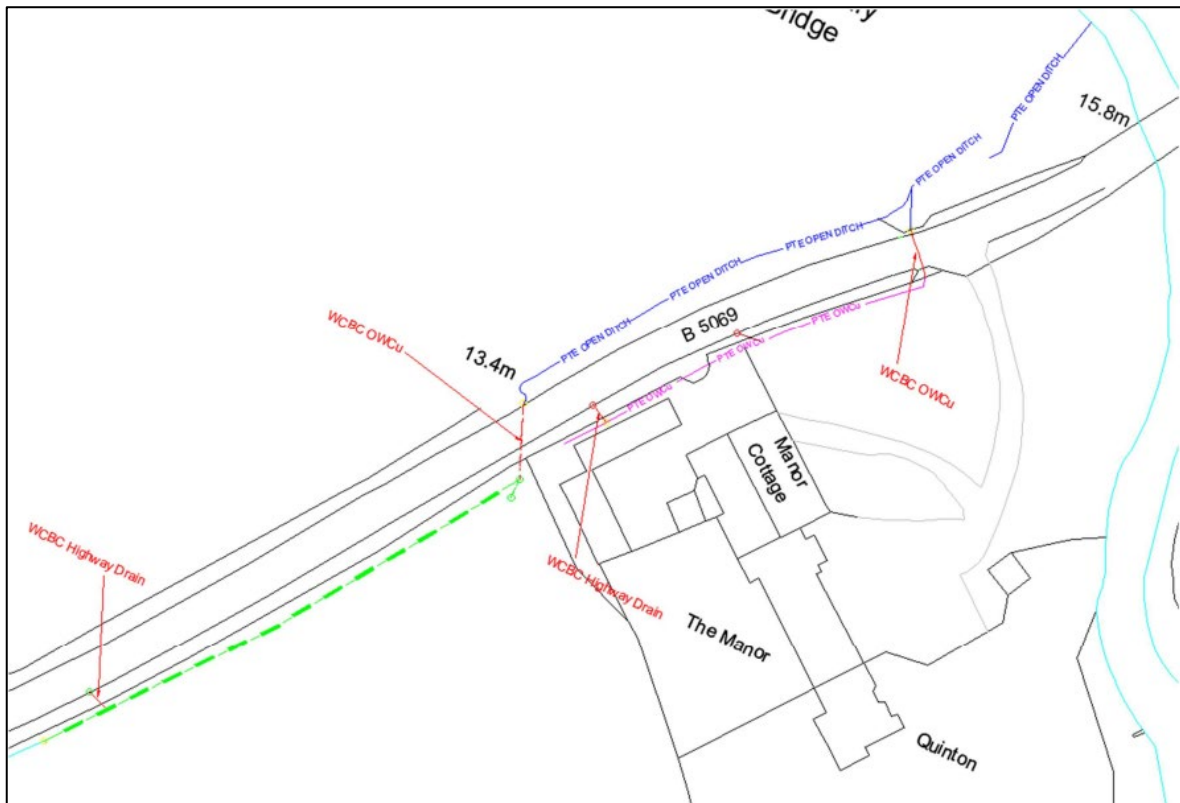


Figure 2: A map of the underground pipework assets in proximity to the B5069 and The Cobbles, Worthenbury.

4. Flood Event Details

Date of Flood Event: 16th October 2024

Date Reported to WCBC: 17th October 2024

Date of Investigation: 18th October 2024

Reason for investigation: Internal flooding of 4 residential properties, extensive flooding of residential gardens and flooding of B5069.

Location: The area to the south of the village of Worthenbury, namely the properties known as Quinton, The Manor, Manor Cottage and The Cobbles.

Event Duration: approximately 9 hours

Flood Duration: approximately 2 days

Identified Source: The flood waters were noted to be from 4 main sources:

- Wych Brook
- Emral Brook
- Worthenbury Brook
- The unnamed ordinary watercourse running alongside the B5069.

4.1 Extent of Flooding:



Figure 3: The fluvial flood risk from the main rivers in Worthenbury as shown by the blue areas. The map is taken from the NRW Flood Risk Maps.



Figure 4: The pluvial flood risk from surface water and ordinary watercourse flooding as shown by the areas in purple. The map is taken from the NRW Flood Risk Maps.

The extent of the flooding largely correlates with that shown in the undefended scenario on the NRW flood maps. The purple areas of surface water flooding indicate where the deeper flood waters collected following the overtopping of the flood embankment.

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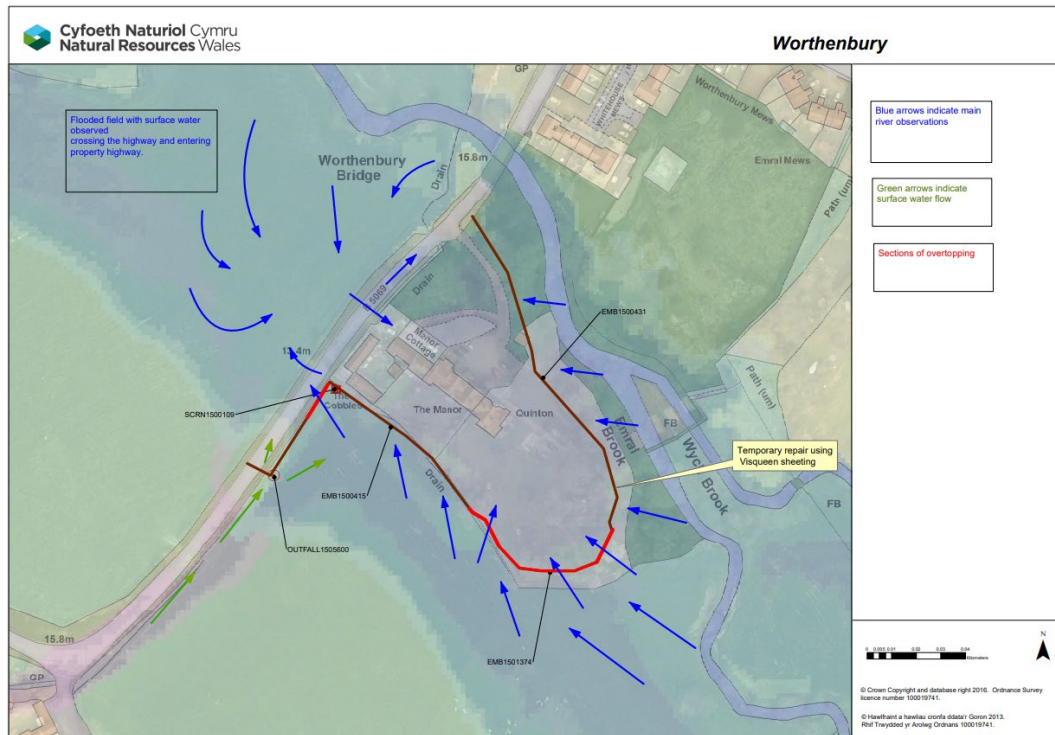


Figure 5: A schematic showing the direction of flooding where blue shows the flooding from main river sources and the green shows the flooding from the ordinary watercourse.

In the initial stages of the flood event, the flood defence is known to have behaved in a typical manner. The flood waters from Wych and Emral Brook left the channel and curved around into the field. In addition to this, pluvial flooding would have occurred in the field from the unnamed ordinary watercourse approaching the defended area from the south west. It is noted that this ordinary watercourse is higher topographically than the highway near to the affected properties. The combination of the multiple flooding sources likely created locally elevated water levels on the south western side of the defence. This resulted in overtopping of the flood embankment in two places: one onto the public and one into the defended area.

It is widely corroborated by the residents that the B5069 flooded first. During the first few hours of the flood event, flooding on the highway can be attributed to two sources. The first of these is from the ordinary watercourse exceeding the channel and flooding the highway from the east. This was exacerbated by the second source of flooding which was the overtopping of the flood embankment. The water on the highway was unable to drain through the gullies and into the culvert that runs within the property boundary of The Cobbles. This was because the flap valve at the outfall of system had been sealed in its closed position by rising water levels in Worthenbury Brook causing flooding to the field in the north. It is unclear if water backed up through the septic tank systems within the defended area at this time, nor if a lack of silt within this system would have had an impact on the flood event. The flood water rose to a level which overtopped the kerbs on the highway. Flood water then entered the defended area through the grounds of The Cobbles and under the wooden fence marking the boundary of the affected properties.

At around 21:00, it was reported that floodwater was entering the embankment under a recent NRW repair. The likely cause of this has been identified as seepage through the root system of a felled tree which was removed.

Further anecdotes from the residents suggest that the floodwater began overtopping the flood embankment to the rear of the properties around 22:00. Given the bowl-like topography inside the flood embankment, this water, combined with that from the highway, pooled and was unable to exit the defence until pumping was commenced by a local farmer.

It is noted that the screen protecting the outlet in the field to the south accumulated a build up debris which was evident when the floodwater receded. It cannot be determined if this a result of suspended debris building up on the screen as the flood level decreased or if it built up during the flood event. Testimony from NRW suggests that this screen was clear before the water level rose in the late afternoon.

In addition to the above, the flap valve in the ownership of WCBC Highways was noted to be heavily vegetated following the flood event. The flap valve is believed to have functioned as designed despite this because the vegetation did not prevent the closure of the valve as the flood water rose in Worthenbury Brook.

4.3 Photos:



Figure 6: A photograph of the overtopping of the flood defence onto the B5069 taken by an NRW officer who attended Worthenbury on 16th October 2024.



Figure 7: A photograph of the seepage under the flood embankment taken by the resident of Quinton on the night of 16th October 2024.



Figure 8: A video screenshot of the overtopping of the flood embankment as taken by a resident on the night of 16th October 2024.



Figure 9: A post-event photograph showing the outfall from the field used as flood storage during the flood event. The screen protecting the outfall is largely blocked due to a buildup of vegetation. The image was taken by a resident of one of the affected properties.



Figure 10: A photograph showing the WCBC outfall at the end of the culvert system taken 12 days post flood event. The photograph was taken by a NRW operative.

4.4 Additional Information:

4.4.1 Drainage Survey

WCBC are in the process of surveying the drainage systems within the area, to obtain clarity of how private

drainage systems interact and influence the highway drainage. The results obtained to date are shown below:

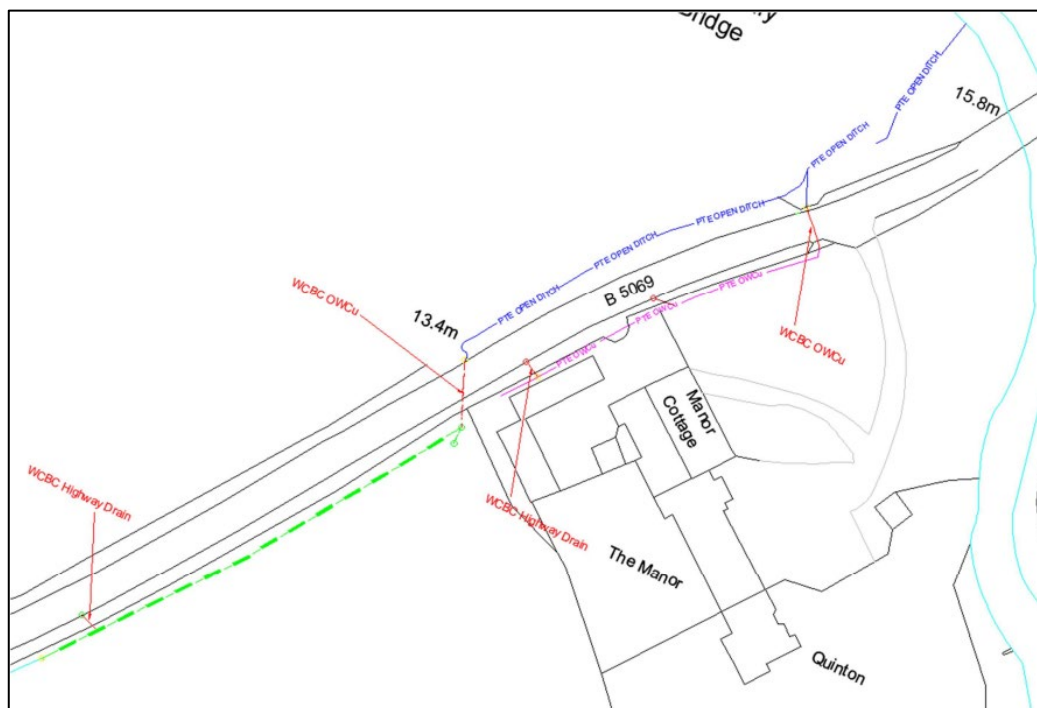


Figure 11: A map of the underground pipework assets in proximity to the B5069 and The Cobbles, Worthenbury. This map was generated as a result of the drainage survey carried out by the WCBC LLFA Team.

During the construction of the flood embankment, the ordinary watercourse was maintained through the overflow system installed by Environment Agency Wales who later passed the responsibility of the system to NRW. There is no indication that the ordinary watercourse still flows through the culverted section within the land of The Cobbles and The Manor. This culvert continues to convey surface water from the highway and the discharge from septic tanks. Given the latter, it remains in private ownership until the culvert crosses under the public highway.

A full and complete survey was unviable due to the amount of silt and water held in the private section of the culvert. As a result, WCBC have been unable to determine the location of the septic tank connections and have relied on information from the residents in the above assumptions.

4.4.2 Catchment Hydrology

Testimony received by both WCBC and NRW suggests that the Worthenbury Brook catchment is becoming flashier over time. This means that the river rises rapidly before receding quickly. WCBC have requested information from other Risk Management Authorities (RMAs) outside the scope of this report, but responses have suggested that no significant changes have occurred in the catchment.

5. Conclusions

In conclusion, the flood event of Wednesday 16th October 2024 was the result of high intensity rainfall over

a number of hours. The levels of water reached such a depth as to cause overtopping of the flood embankment into the defended area and on to the public highway. This exacerbated flooding on the highway, as outfalls were compromised. As a result, flood water entered the defended area to both the north west and the south east. This then caused significant flooding to four properties which lasted for approximately two days. The flood water had to be removed from the defended area using a pump which was provided by a local landowner.